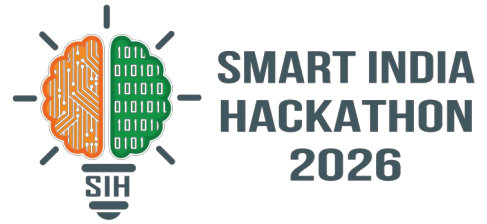


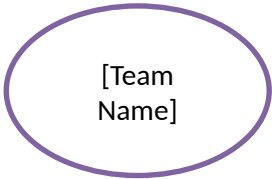
SMART INDIA HACKATHON 2026

SatQuery AI

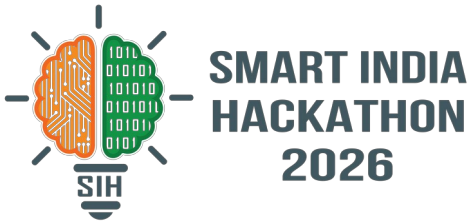


- Problem Statement ID – 26167
- Problem Statement Title – SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- Theme – Space Technology
- PS Category – Software
- Team ID – *to be filled*
- Team Name – *[Team Name]*





SatQuery AI — Ask Satellite Imagery a Question



Proposed Solution (Describe your Idea/Solution/Prototype)

THE PROBLEM

- **No interface accepts a question** — every remote-sensing tool is single-task and needs a GIS expert.
- **One image is often not enough** — cloud blocks optical sensors; change needs two dates.
- **Generic AI chatbots invent numbers** and cannot show why they answered.

OUR SOLUTION — HOW IT ADDRESSES IT

- Upload **GeoTIFF imagery** — one image, an **optical + SAR pair**, or **two dates** — and ask in plain English.
- An **agentic router** classifies the task, **checks the inputs**, and runs the right specialist from a **fixed tool registry**.
- Returns an **answer + map evidence (lat/long) + calibrated confidence + execution trace + report**.
- Specialists for **VQA, grounding, change analysis and optical-SAR fusion**; adapted AI model: *see model section*.

INNOVATION & UNIQUENESS

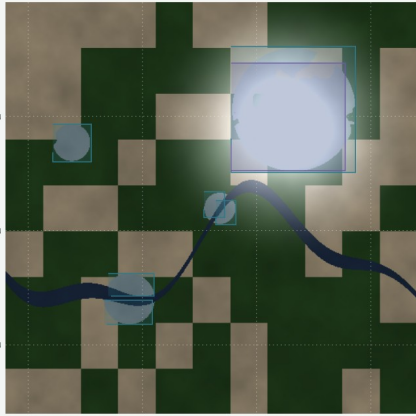
**Refuses the impossible**
No model runs if the uploads cannot answer the question

**Facts before words**
Answer written only from measured evidence — never invented

**Sees under clouds**
Radar + optical fusion measures what the camera misses

**Auditable & offline**
Trace of exactly what ran; no internet needed

PROTOTYPE — WORKING TODAY



CROSS-MODEL
optical_sar - classical - brows...
0.93
CONFIDENCE

SAR detects 6 built-up areas totalling 2815.3 ha. Cloud obscures 6% of the optical scene. 1 of them — 1607.4 ha — lies beneath that cloud and is invisible to the optical sensor. Radar recovered it because radar penetrates cloud and built structures return strongly from corner reflection. That is information neither sensor provides alone.

co-registration verified 0 px
Zused - fusion_late_v000.1.0 0.95 PASS

EXECUTION TRACE router rules - engine classical

- Input validated 0.2 ms
 - data - optical_sar
- Task identified 0.9 ms
 - data - routers
- Compatibility check 1.0 ms
 - cross_model - matched /optical and/
 - data - confidence, router
- Tool selected 1.0 ms
 - optical_sar satisfies optical_sar
 - optical_sar (fusion_late_v0)
 - data - registry_size

Parameters

Query: “Use the optical and SAR images together to identify built-up regions.”

SAR finds **2,815 ha** built-up; **1,607 ha** of it lies **under cloud**, invisible to the optical sensor.

Confidence **0.93** · every step in the trace.

Synthetic demo scene — disclose on the slide

Also live: refusal with no model run · two-date change detection · GeoJSON + report export · real GeoTIFF upload.

TECHNICAL APPROACH

[Team
Name]

Technologies to be used



Backend

Python · NumPy · Pillow · FastAPI



Web app

React 19 · TypeScript · Vite · Tailwind



3D + maps

three.js sensor stack · GeoJSON (EPSG:4326)



Geospatial

GeoTIFF / TIFF · geotiff.js · CRS checks



Image analysis

Lee filter · Otsu · NDVI/NDWI · change vector analysis

AI model & training

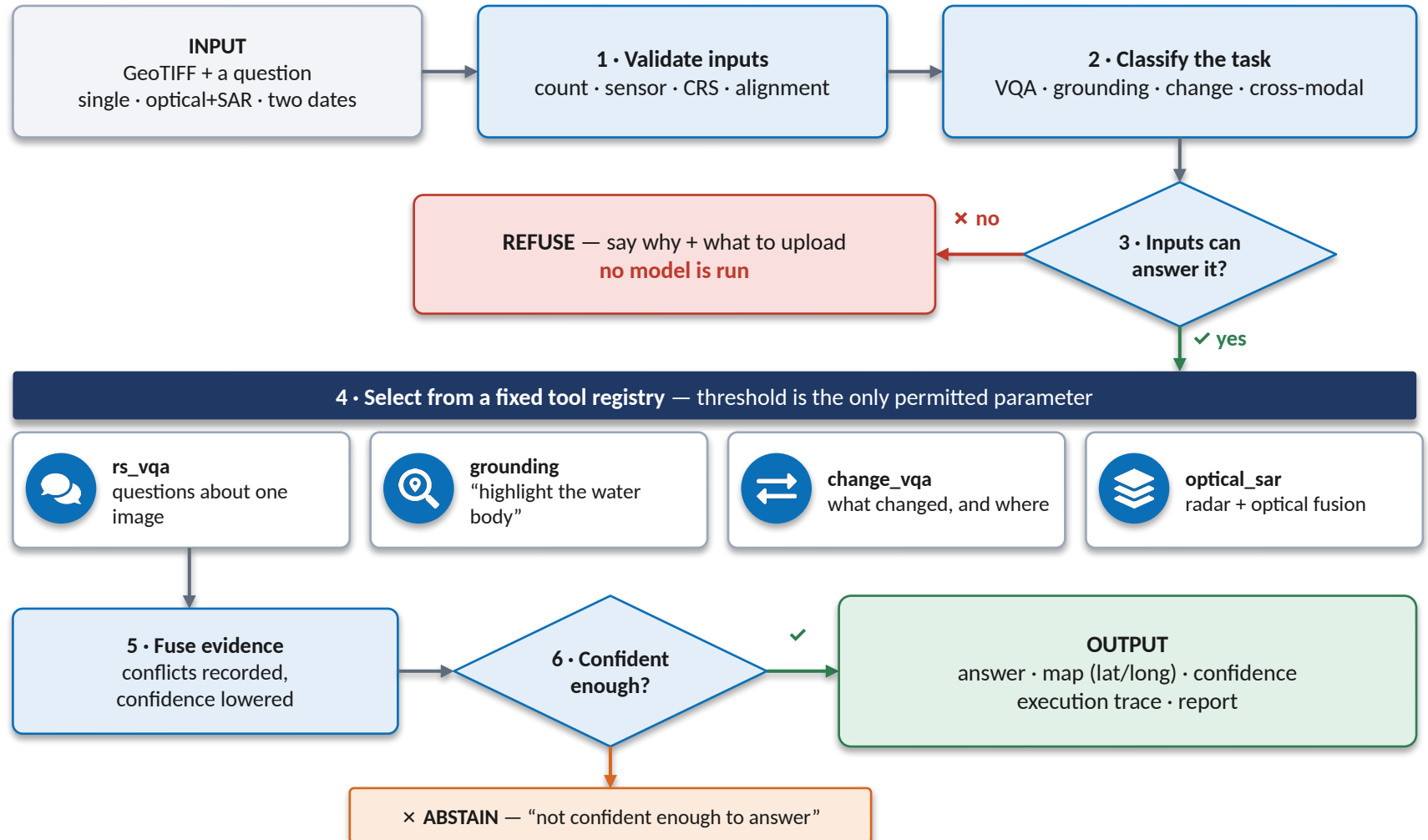
Fill from the model document

Prototype: [link](#)

Demo video: [YouTube link](#)

Code: [GitHub link](#)

Methodology and process for implementation



FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
Technical — built & running	AI model not yet connected to every specialist	Model “socket” built and tested; classical path answers today and the trace labels which engine ran
Data — robust	ISRO/SAC test set is hidden; Cartosat/RISAT data is not public	Input validation, calibrated confidence and abstention — built for data we have never seen
Trust — verifiable	AI answers that sound right but are wrong	Refuse before any model runs ; answer phrased only from measured evidence; full trace
Operational — demo-safe	Venue Wi-Fi, limited GPU	Fully offline ; only the model runtime needs a GPU; pre-computed demo results clearly labelled

MEASURED TODAY — on synthetic test scenes

1.00

Water grounding IoU

0.996

Vegetation IoU

0.945

SAR built-up F1

0.770

Change detection F1

0.033

Calibration error (ECE)

~90 ms

Cross-modal query

VIABILITY

**Technically viable** — open-source stack; every capability measured, not assumed**Economically feasible** — free open datasets and tools; runs on a laptop**Operationally practical** — browser app, drag-and-drop GeoTIFF, downloadable report

IMPACT AND BENEFITS

Potential impact on the target audience

-  **District & disaster officers** — flood, damage and construction questions without a GIS expert
-  **Agriculture & water departments** — vegetation and water-body extent from imagery
-  **Urban planners** — built-up growth between two dates, mapped
-  **ISRO / SAC analysts** — routine questions faster, with an auditable trace
-  **Students & researchers** — remote sensing without specialist software

Benefits of the solution

-  **Social** — faster, evidence-backed answers in disasters; wider access to space data
-  **Economic** — less specialist time per question; open-source, no licence cost
-  **Environmental** — monitor vegetation, water and land-use change over time
-  **Governance** — every answer traceable to a measurement — fit for official use

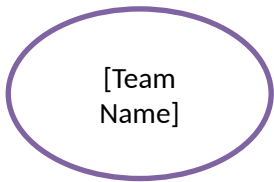
Capability	Single-task RS tool	Generic AI chatbot	SatQuery AI
Ask in plain language	×	✓	✓
Picks the right analysis automatically	×	×	✓
Checks inputs, refuses impossible questions	×	×	✓
Optical + SAR together (sees under cloud)	⦿	×	✓
Every number traceable to a measurement	✓	×	✓



Works where the camera can't

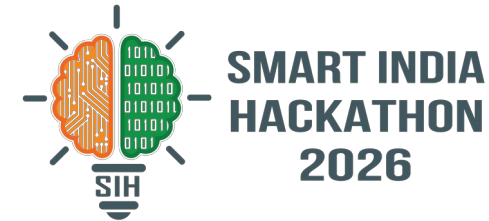
Radar sees through cloud and at night. Fusing it with optical turns a cloudy scene into a measured answer.

Add one real-world number here, with its source



[Team Name]

RESEARCH AND REFERENCES



Details / Links of the reference and research work

Aspect	Conventional RS tools / generic VLMs	SatQuery AI
Interface	GIS software, expert-only	Plain-language question
Model choice	Manual, one task per tool	Automatic, from a fixed registry
Input checking	Assumed	Validated — impossible questions refused
Sensors & time	One image, one sensor	Optical + SAR fusion; two-date change
Explainability	Free text or a score	Evidence + confidence + execution trace

References

- [1] SIH 2026, PS26167 “SatQuery AI”, ISRO — official problem statement.
- [2] BigEarthNet.txt — named in the PS: arxiv.org/abs/2603.29630
- [3] VRSBench, remote-sensing vision-language benchmark — huggingface.co/datasets/xiang709/VRSBench
- [4] S. Lobry et al., “RSVQA: Visual Question Answering for Remote Sensing Data,” IEEE TGRS, 2020.
- [5] CDVQA, change-detection visual question answering benchmark — named in the PS.
- [6] J.-S. Lee, “Digital image enhancement and noise filtering by use of local statistics,” IEEE TPAMI, 1980.
- [7] N. Otsu, “A threshold selection method from gray-level histograms,” IEEE Trans. SMC, 1979.

Add model papers from the model document

PROJECT RESOURCES

REFUSED · NO MODEL INVOKED

no tool · classical · browser p...

0.00

CONFIDENCE

This question needs a bi-temporal pair, but a co-registered optical and SAR pair were supplied.

WHAT TO DO

Upload an image for T1 and an image for T2 — two dates of the same area.

Load the bi-temporal pair and re-ask

Prototype: [link](#)

Demo video: [link](#)

GitHub: [link](#)

Refusal demo — no model invoked

DATASETS NAMED IN THE PS

BigEarthNet.txt
Sentinel-1 SAR + Sentinel-2 optical, with text — adaptation

VRSBench · RSVQA
single-image VQA, grounding, captioning — evaluation

CDVQA
change-based question answering — evaluation

ISRO/SAC set
Cartosat-2S + RISAT pairs, answers hidden — final scoring